25900685

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular Examination – October 2025

Third Semester

Discipline Specific Core Course (DSC) – MG3DSCZGY203

BIOLOGICAL BASIS OF BEHAVIOR – III

(2024 Admission Onwards)

Scheme for valuation

Duration: 2 Hours **Maximum Marks:** 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap).

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A – Multiple Choice Questions

(Answer all questions. Each question carries 1 mark.) **[1×10 = 10 marks]**

1. **Which of the following best describes homeostasis?** **[K]/[CO1]**
   b) The process of maintaining internal balance despite external changes
2. **The nerve that carries smell sensation to the brain is:** **[K]/[CO1,CO2]**
   b) Olfactory nerve
3. **Bowman's glands in the olfactory epithelium secrete:** **[U]/[CO1,CO2]**
   b) Mucus and odorant-binding proteins
4. **Which of the following statements is TRUE regarding Charles Bonnet Syndrome?** **[U]/[CO3]**
   d) Patients usually maintain normal cognitive function
5. **Which part of the brain processes temperature sensations?** **[U]/[CO3]**
   c) Somatosensory cortex
6. **Hypermetropia can be corrected by using:** **[U]/[CO3]**
   b) Convex lens
7. **Which of the following hormones has a direct effect on blood pressure?** **[U]/[CO4]**
   b) Vasopressin
8. **Dysregulation of the HPA axis can lead to:** **[U]/[CO4]**
   d) Reduced stress management
9. **Postpartum depression is classified under:** **[U]/[CO4]**
   b) Mood disorders
10. **Dysregulation of which axis is commonly associated with anxiety disorders?** **[U]/[CO4]**
    c) HPA axis

Part B – Short Answer Questions

(Answer any 10 questions. Each question carries 2 marks.) **[2×10 = 20 marks]**

11. **State the cranial nerves involved in carrying taste signals.** **[U]/[CO2]**

- Facial nerve (VII) – anterior 2/3 of tongue
- Glossopharyngeal nerve (IX) – posterior ⅓
- Vagus nerve (X) – epiglottis and pharynx *(Mention the three cranial nerves only - give full marks)*

12. Enumerate the functions of taste buds. [U]/[CO1,CO2]

- Detects different taste sensations – sweet, sour, salty, bitter, and umami.
- Help in identifying safe and harmful foods.
- Send taste signals to the brain through cranial nerves.
- Stimulate secretion of saliva and digestive juices.
- Contribute to the overall flavor perception along with smell.
- Aid in maintaining nutritional balance by promoting intake of essential nutrients. *(Any two points- 1 mark each)*

13. State the function of rod cells. [U]/[CO3]

- Rod cells are photoreceptor cells in the retina of the eye that are highly sensitive to dim light and are responsible for black-and-white (night) vision.

1. Enable vision in dim light (night or scotopic vision).
2. Detects shape, movement, and brightness but not color.
3. Contain the pigment rhodopsin, which helps in light detection.
4. Provide peripheral vision.
5. Function best under low light intensity. *(Any two functions - 1 mark each)*

14. Which is the layer of epidermis present in thick skin and comment on it? [U]/[CO3]

- The layer of epidermis present only in thick skin is the Stratum lucidum.
- It is a thin, clear, transparent layer found between the stratum granulosum and stratum corneum.
- Present only in palms of the hands and soles of the feet.
- Made up of dead, flattened keratinocytes containing eleidin, a translucent protein that helps protect underlying layers.
- Provides extra protection and reduces friction in areas exposed to constant pressure.*(Stratum lucidum- 1 mark, any one comment- 1 mark)*

15. Give the pathway of visual impulses from the retina to the visual cortex. [U]/[CO3]

- Retina (Rods & Cones) → Bipolar Cells → Ganglion Cells → Optic Nerve → Optic Chiasma → Optic Tract → Lateral Geniculate Body (Thalamus) → Optic Radiations → Visual Cortex (Occipital Lobe). (***NB****: A clear, precise flowchart showing the sequence is sufficient for full marks; no diagram or detailed explanation is needed.)*

16. Describe the function of the middle ear. [U]/[CO1,CO2]

- Transmission of sound: Transfers sound vibrations from the tympanic membrane (eardrum) to the inner ear through the ossicles — malleus, incus, and stapes.
- Amplification: The ossicles amplify the sound vibrations to overcome the resistance of the fluid in the inner ear.
- Pressure equalization: The Eustachian tube connects the middle ear to the nasopharynx and helps equalize air pressure on both sides of the eardrum.
- Protection: The tensor tympani and stapedius muscles protect the inner ear from loud sounds by dampening excessive vibrations. *(Any 2 points: 1 mark each)*

17. Mention the ordered sequence of auditory relay stations from periphery to cortex. [U]/[CO2]

- Cochlea → Cochlear Nucleus → Superior Olivary Complex → Inferior Colliculus → Medial Geniculate Body (Thalamus) → Auditory Cortex .(***NB****: A clear, precise flowchart showing the sequence is sufficient for full marks; no diagram or detailed explanation is needed.)*

18. A woman with PCOD complains of infertility. How do assisted reproductive techniques apply in her management? [A]/[CO4]

- Ovulation induction therapy – Use of drugs like *Clomiphene citrate*, *Letrozole*, or *Gonadotropins* to stimulate ovulation.

- Intrauterine insemination (IUI) – Washed sperm are directly introduced into the uterus near the time of ovulation.
- In vitro fertilization (IVF) – Eggs are collected, fertilized in the lab, and the resulting embryos are transferred to the uterus.
- Intracytoplasmic sperm injection (ICSI) – A single sperm is injected directly into an egg; used when sperm count or motility is low.
- Laparoscopic ovarian drilling (LOD) – A surgical technique to destroy androgen-producing tissue, improving ovulation.
- Cryopreservation of embryos or oocytes – Useful for preserving fertility if repeated cycles are needed.
- Lifestyle modification support – Though not ART itself, weight reduction and insulin-sensitizing drugs like *metformin* enhance ART success. *(Any two methods with explanations or four methods mentioned briefly can be given full marks.)*

19. Mention two common physical symptoms of premenstrual dysphoric disorder. [K]/[CO4]

- Abdominal bloating, breast tenderness or swelling, headache or migraine, muscle or joint pain, fatigue, changes in appetite, sleep disturbances, weight gain, back pain, and swelling of hands or feet.(*Any two symptoms - 1 mark each.)*

20. Why is the pituitary gland called the "master gland"? [K]/[CO4]

- Because it controls and regulates the activity of other endocrine glands such as the thyroid, adrenal, and reproductive glands by secreting hormones that influence their functions.*(Give full marks for relevant explanation)*

21. Write short notes on Graves' disease. [U]/[CO4]

- Graves' disease is an autoimmune disorder causing hyperthyroidism.
- The body produces TSH receptor antibodies that overstimulate the thyroid gland.
- Leads to excess secretion of thyroid hormones (T_3 and T_4).
- Symptoms: weight loss, heat intolerance, rapid heartbeat, tremors, anxiety, and exophthalmos (bulging eyes).
- Treatment: antithyroid drugs, radioactive iodine therapy, or partial thyroidectomy. ***(Liberal valuation)***

22. Name one mood stabilizer used in the treatment of bipolar disorder. [K]/[CO4]

- Lithium – the most common and effective traditional mood stabilizer
- Valproate (Valproic acid) – controls manic episodes
- Carbamazepine – used when lithium is not tolerated
- Lamotrigine – effective for preventing depressive episodes
- Atypical antipsychotics – such as olanzapine, quetiapine, and risperidone for mood stabilization
- Psychotherapy – cognitive behavioral therapy (CBT) helps manage mood swings
- Mindfulness and meditation – reduce stress and emotional instability
- Regular sleep schedule – stabilizes mood and prevents manic/depressive triggers
- Physical exercise – boosts endorphins and supports emotional balance
- Avoidance of alcohol and recreational drugs – prevents worsening of symptoms
- Support groups and family therapy – improve emotional regulation and adherence to treatment. *(Any one stabilizer - 2 mark or a general description can give full marks- Liberal valuation)*

Part C – Short Essay Questions

(Answer any 5 questions. Each question carries 4 marks.) **[4×5 = 20 marks]**

23. Describe the functional importance of olfaction in daily life activities. [U]/[CO1]

- Detection of hazards: Helps identify dangers such as smoke, gas leaks, or spoiled food.
- Enhancement of taste: Works with taste buds to create the perception of flavor.
- Emotional connection: Certain smells evoke memories and emotions.

- Social communication: Body odor and perfumes influence social interactions and attraction.
- Appetite regulation: Pleasant food odors stimulate appetite, while unpleasant ones suppress it.
- Environmental awareness: Helps recognize surroundings and personal hygiene. *(Explanation of any 4 points- 4 marks or a good account on functional importance - 4 marks)*

24. Explain the different primary taste sensations and give examples. [U]/[CO2]

- Sweet: Caused by sugars and some amino acids — e.g., candy, honey, fruits.
- Sour: Produced by acids — e.g., lemon, vinegar.
- Salty: Due to salts, mainly sodium ions — e.g., table salt, chips.
- Bitter: Often from alkaloids — e.g., coffee, cocoa, neem.
- Umami: A savory taste from glutamate and certain amino acids — e.g., meat, cheese, soy sauce.

(Mentioning all five primary tastes and example :2 mark, Brief explanation:2 marks)

25. Explain how nociceptors are involved in the perception of pain through the skin. [U]/[CO3]

- Nociceptors are specialized sensory receptors in the skin that detect painful or damaging stimuli such as extreme heat, pressure, or chemicals.
- When activated, they generate nerve impulses that travel through sensory neurons to the spinal cord.
- These impulses are then relayed to the brain, particularly the thalamus and somatosensory cortex, where pain is perceived.
- This mechanism helps the body recognize and respond to potential harm, promoting protective actions like withdrawal from the source of injury. *(definition:1 mark, Comment: 3 marks or a good account on nociceptors: 4 marks)*

26. Compare Meissner's and Pacinian corpuscles. [U]/[CO3]

Feature	**Meissner's Corpuscles**	**Pacinian Corpuscles**
Location	Present in dermal papillae of hairless (glabrous) skin such as fingertips and lips	Located deep in the dermis and subcutaneous tissue
Stimulus Detected	Light touch and texture changes	Deep pressure and vibration
Adaptation Speed	Rapidly adapting	Very rapidly adapting
Structure	Small, oval, encapsulated receptors	Large, onion-like layered structure

(Any two comparison : 2 mark each)

27. Describe the important functions of the endocrine system in our body. [U]/[CO4]

- Regulates metabolism: Controls energy production and nutrient use through hormones like insulin and thyroxine.
- Maintains homeostasis: Keeps internal conditions stable, including blood pressure, temperature, and water balance.
- Controls growth and development: Growth hormone and thyroid hormones promote body growth and tissue development.
- Regulates reproduction: Hormones like estrogen, progesterone, and testosterone control sexual development and fertility.
- Coordinates stress response: Adrenal hormones such as cortisol and adrenaline help the body respond to stress.
- Regulates mood and behavior: Hormones influence emotions, sleep, and mental well-being.
- Regulates blood glucose levels: Insulin and glucagon maintain normal blood sugar balance.

- Controls calcium and bone metabolism: Parathyroid hormone and calcitonin regulate calcium levels and bone strength.
- Influences circadian rhythm: Melatonin from the pineal gland controls sleep–wake cycles.
- Aids in digestion: Hormones like gastrin, secretin, and cholecystokinin regulate digestive processes.
- Supports immune function: Thymosin promotes immune cell development.
- Regulates blood pressure and fluid balance: Aldosterone and antidiuretic hormone (ADH) maintain water and salt balance.
- Promotes cell repair and regeneration: Growth factors assist in tissue maintenance and healing.
- Integrates with the nervous system: Works with the hypothalamus to coordinate body responses to internal and external changes**.** *(Explain any 4 points with brief explanations – 4 marks or mention any correct 6 points- 4 marks)*

28. Explain the interaction between the sympathetic nervous system and adrenal medulla in stress.

[U]/[CO4]

- During stress, the sympathetic nervous system (SNS) is activated as part of the "fight or flight" response.
- The SNS stimulates the adrenal medulla (inner part of the adrenal gland) through sympathetic nerve fibers.
- The adrenal medulla then releases epinephrine (adrenaline) and norepinephrine (noradrenaline) into the bloodstream.
- These hormones increase heart rate, blood pressure, respiration, and glucose release, preparing the body to handle stress.
- This shows a close coordination where the SNS provides rapid nerve stimulation, and the adrenal medulla ensures a prolonged hormonal effect to manage stress efficiently. *(Give full marks for relevant explanation)*

29. Discuss how hormonal changes in the ovaries prepare the female body for fertilization and implantation. [U]/[CO4]

- Follicular phase: Follicle-stimulating hormone (FSH) promotes the growth of ovarian follicles, which secrete estrogen. Estrogen thickens the endometrium and prepares the cervix for sperm entry.
- Ovulation: A surge in luteinizing hormone (LH) triggers the release of an egg (ovum) from the ovary.
- Luteal phase: After ovulation, the corpus luteum forms and secretes progesterone and some estrogen.
- Progesterone stabilizes and further thickens the endometrium, making it suitable for implantation.
- If fertilization occurs, progesterone continues to support the uterine lining; if not, hormone levels drop, leading to menstruation. (*1 mark each for mentioning each phase (follicular, ovulation, luteal, and outcome) along with the specific hormone involved and its function.)*